

SMART AGRICULTURE DRONE USING ESP32

Abstract

The Smart Agriculture Drone project is designed to demonstrate the use of IoT and automation in agriculture. The system uses an ESP32 microcontroller to monitor environmental conditions such as temperature, humidity, soil moisture, and crop distance. Based on the collected data, the system automatically activates a spraying mechanism when the soil is dry and the crop is within the specified range. This project helps in reducing water and pesticide wastage while improving farming efficiency.

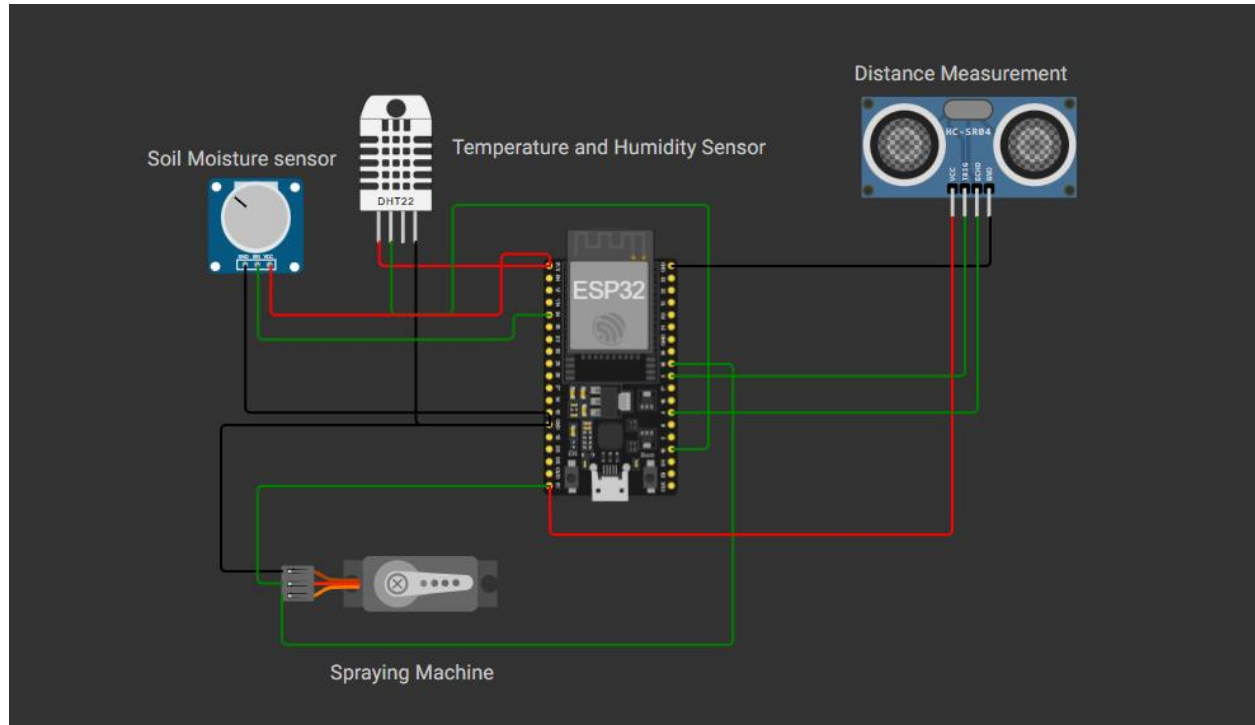
Objectives

- To monitor temperature and humidity.
 - To simulate soil moisture measurement.
 - To detect crop distance using an ultrasonic sensor.
 - To automate the spraying process.
 - To demonstrate smart farming using ESP32.
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Components Used

1. ESP32 Development Board
 2. DHT22 Temperature and Humidity Sensor
 3. HC-SR04 Ultrasonic Sensor
 4. Servo Motor
 5. Potentiometer (Soil Moisture Simulation)
 6. Jumper Wires
 7. Wokwi Simulator
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Circuit Diagram



Working Principle

The ESP32 continuously reads data from the sensors.

- DHT22 measures temperature and humidity.
- Potentiometer simulates soil moisture values.
- HC-SR04 ultrasonic sensor measures the distance from the crop.
- Servo motor acts as the spraying mechanism.

The ESP32 checks the following condition:

- Soil value less than 1500
- Crop distance less than 50 cm

If both conditions are satisfied, the servo motor rotates to 90 degrees and spraying starts. Otherwise, spraying remains OFF.

Algorithm

1. Start the system.
 2. Read temperature and humidity from DHT22.
 3. Read soil moisture value from potentiometer.
 4. Measure crop distance using HC-SR04.
 5. Check threshold conditions.
 6. If soil is dry and crop is nearby, activate spraying.
 7. Display values on Serial Monitor.
 8. Repeat the process continuously.
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Source Code

```
#include <DHT.h>

#include <ESP32Servo.h>

#define DHTPIN 15

#define DHTTYPE DHT22

#define SOIL_PIN 34

#define TRIG 5

#define ECHO 4

Servo sprayMotor;

DHT dht(DHTPIN, DHTTYPE);


void setup() {

    Serial.begin(115200);

    dht.begin(); // start DHT sensor

    pinMode(TRIG, OUTPUT);

    pinMode(ECHO, INPUT);

    sprayMotor.attach(18); // servo pin

    sprayMotor.write(0); // spray OFF
```

```
}
```

```
void loop() {
```

```
    // sensor readings
```

```
    float temperature = dht.readTemperature();
```

```
    float humidity = dht.readHumidity();
```

```
    int soilValue = analogRead(SOIL_PIN);
```

```
    // ultrasonic sensor
```

```
    digitalWrite(TRIG, LOW);
```

```
    delayMicroseconds(2);
```

```
    digitalWrite(TRIG, HIGH);
```

```
    delayMicroseconds(10);
```

```
    digitalWrite(TRIG, LOW);
```

```
    long duration = pulseIn(ECHO, HIGH);
```

```
    float cropDistance = duration * 0.034 / 2;
```

```
    // display data
```

```
    Serial.print("Temperature = ");
```

```
    Serial.println(temperature);
```

```
    Serial.print("Humidity = ");
```

```
    Serial.println(humidity);
```

```
    Serial.print("Soil = ");
```

```
    Serial.println(soilValue);
```

```
    Serial.print("Distance = ");
```

```
    Serial.println(cropDistance);
```

```
// spray condition
if (soilValue < 1500 && cropDistance < 50)
{
    sprayMotor.write(90); // start spray
    Serial.println("Spraying ON");
}
else
{
    sprayMotor.write(0); // stop spray
    Serial.println("Spraying OFF");
}
delay(2000);
}
```

Results

The project successfully monitored environmental conditions and controlled the spraying mechanism automatically.

Sample Output

Temperature = 29°C

Humidity = 70%

Soil Value = 1200

Distance = 35 cm

Spraying ON

Advantages

- Reduces manual effort.
 - Saves water and pesticides.
 - Improves farming efficiency.
 - Supports smart agriculture concepts.
 - Easy to implement and expand.
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Future Scope

- GPS-based navigation.
 - Camera-based crop monitoring.
 - Mobile app integration.
 - Cloud data storage.
 - AI-based disease detection.
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Conclusion

The Smart Agriculture Drone project successfully demonstrates the use of IoT and automation in agriculture. The ESP32 processes data from multiple sensors and automatically controls the spraying mechanism based on predefined conditions. This project provides a simple and effective solution for smart farming applications.